Nikhil Verma

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Education

Birla Institute of Technology and Science *Pilani, Rajasthan, IN*

B.E. (HONS.) ELECTRICAL AND ELECTRONICS ENGINEERING *Aug 2014 - Jul 2018*

Cummulative GPA 8.01

Publications

Neural Conversational QA: Learning to Reason vs Exploiting Patterns *Under Review*

Nikhil Verma, Abhishek Sharma, Dhiraj Madan, Danish Contractor, Harshit Kumar and Sachindra Joshi
Paper under review at the 2020 Conference on Empirical Methods in Natural Language Processing.

Generating Dialog System Workspaces *App 16/892805*

Danish Contractor, Nikhil Verma, Harshit Kumar and Sachindra Joshi
Patent filed with the US Patent and Trademark Office.

Experience

IBM Research *New Delhi, IN*

STAFF RESEARCH ENGINEER *Jul 2018 - Present*

- Leading efforts to associate chat conversations with historical events as part of the Watson AIOps team.
- Developed UrcaNET Ensemble, one of the best performing models on the ShARC leaderboard.
- Exposed spurious patterns in the ShARC dataset that were being exploited by the top performing models.
- Developed IRLQA, a PyTorch framework to develop models for question answering on the QuAC task.

Microsoft Research *Bangalore, IN*

RESEARCH INTERN *Jan 2018 - Jul 2018*

- Worked towards developing novel methods to generate natural language summaries describing source code.
- Conceptualized and implemented a weighted attention mechanism using OpenNMT (PyTorch).
- Our model was able to better focus on rare tokens producing improved NL descriptions for `bash` commands.

IBM Research *New Delhi, IN*

RESEARCH INTERN *May - Jul 2017*

- Improved techniques for generating answerable questions from a given source text.
- Implemented a novel approach using a syntactic sequence to sequence model to handle the problem of unknown sequences.
- Achieved improvements in automatic evaluation metrics using a much smaller vocabulary.

BITS Pilani *Pilani, Rajasthan, IN*

TEACHING ASSISTANT *Aug - Dec 2017*

- Introduced Numpy based programming assignments in the most sought after course for Neural Networks on campus.
- Managed the logistics of online assignments with over 180 people attempting the test at the same time.

Department of Visual Media *Pilani, Rajasthan, IN*

BACKEND DEVELOPER, COORDINATOR *Aug 2014 - May 2016*

- Developed the backend for the official websites of all three major college festivals.
- Head of the department of over 40 skilled enthusiasts.
- Our systems managed over 1000 participants over the course of four days.

Skills

Proficient Python, NumPy, PyTorch, and Django**;** Recurrent Neural Networks, Seq2Seq Networks, Attention Mechanisms.
Experienced JavaScript, C/C++, Java, Bash, LaTeX**;** TensorFlow, Keras, Flask, OpenNMT**;** MySQL, SQLite, MongoDB, and Redis

Projects

Watson AIOps
Link

HELP RESOLVE IT INCIDENTS WITH SMARTER INSIGHTS — *Late 2019*

- Leading efforts to analyze and associate chat conversations with key events that have happened in the past.
- Part of the team which developed an orchestrator to coordinate between various microservices and SLACK.

ShARC
Link

EXPOSING PATTERNS IN WIDELY USED DATASETS — *Early 2020*

- Assisted in finding spurious patterns in the ShARC dataset which were being exploited by various state of the art models.
- Proposed a modified version of the dataset with recommendations on how to discourage models from relying on such patterns.
- Developed UrcaNET Ensemble, one of the best performing models on the ShARC leaderboard which addresses the issue.

IRLQA

CONVERSATIONAL QUESTION ANSWERING FRAMEWORK — *Early 2019*

- Developed IRLQA, a PyTorch based framework for training and evaluating models for conversational question answering.
- Implemented state-of-the-art models on the QUAC task thereby severely reducing the dependency on ALLENNLP.
- Developed a PASSAGE ATTENTION FLOW mechanism, a novel improvement over existing BiDAF based models.

Bash2NL
Report

EXPLAIN SOURCE CODE IN NATURAL LANGUAGE — *April 2018*

- Used sequence to sequence networks to describe `bash` commands in natural language.
- Conceptualized and implemented supervised and weighted attention mechanisms in OPENNMT-PY.
- Our model is able to focus more on rarely occurring words, thus generating better descriptions.

Neural Question Generation
Report

A STEP CLOSER TO CREATING FAQS — *June 2017*

- Used sequence to sequence models for generating answerable questions from a given sentence.
- Experimented with both, the traditional RNN based seq2seq models as well as a fully convolutional model.
- Used a combination of reusable `<unk>` tokens and lemmatization to tackle the problem of unknown sequences.
- Achieved improvements in automatic evaluation metrics using a much smaller vocabulary.

Predicting Code Switch Points
Report | Code

BECAUSE "KABHI KABHI" WE SWITCH — *Oct 2017*

- Developed a novel method to predict possible code-switch points given a monolingual sentence.
- The model is based on stacked, bidirectional LSTMs containing character-level and word-level embeddings.
- The system only requires word-level language tagged data for training unlike other methods which require extensive preprocessing.

Autonomous Mobile Bots
Report | Code

A TOY IMPLEMENTATION USING DEEP Q LEARNING — *Apr 2017*

- Trained a three layer deep reinforcement learning network to steer a bot automatically given a path.
- Created OpenAI-Gym style APIs as an interface between V-Rep and Keras to facilitate training.
- Made an actual hardware implementation of the model based on Raspberry Pi.

Presidential Poll Predictor
Code

SENTIMENT ANALYSIS USING CNNS — *Apr 2016*

- Used CNNs to predict the US presidential elections by performing sentiment analysis on real time twitter feed.
- Implemented CNNs in Python from scratch, without the use of autodifferentiation libraries.

Event Management Systems
Code

END TO END MANAGEMENT FOR COLLEGE FESTS — *Feb 2016*

- Developed the centralised backend for the official websites of all college fests.
- The system is used to manage registration, billing, accomodation and event participation of over 1000 participants during the fests.

Interests

While I cook and play the guitar every once in a while, I'm constantly trying to check items off my reading list.

In a previous life, I also used to write at nikhilweee.me about all sorts of things - work or play.